Marshal Turner

Machine Learning Engineer | NLP / LLM Systems
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Machine Learning Engineer with 4+ years of hands-on experience building, deploying, and scaling production-grade ML models that deliver multimillion-dollar business impact. Experienced with end-to-end model development using Python, Scikit-learn, PyTorch (via advanced RL projects), and cloud tools (AWS, Snowflake). Proven success with high-precision classifiers (91% precision / 89% recall), propensity models tied to $4M+ annual revenue, and forecasting systems reducing error to 1.9% MAPE, alongside automating workflows that cut manual effort by over 50%. Currently pursuing a Master's in Analytics at Georgia Tech to deepen expertise in advanced deep learning, MLOps, and scalable AI systems. Passionate about architecting reliable, performant ML solutions that drive measurable outcomes.

Technical Skills

Languages & Frameworks: R, Python, Scala, Spark, SQL (T-SQL), JavaScript
Libraries: Pandas, Selenium, Matplotlib, Seaborn, Scikit-learn, Tensorflow, Pytorch, Langchain. Keras
ML & DLs: PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, PPO, QMIX, CNNs, GRUs
MLOps & Cloud: Docker, GitHub, GitLab, AWS, Azure, Snowflake, Boomi
Data & Visualization: Pandas, Matplotlib, Seaborn, Tableau, Power BI, Microsoft Report Builder, D3, Qlik
Other: Selenium, Postman, Lucidchart, JIRA REST API, Django, ARIMA, HistGradientBoosting

Professional Experience

Flexential Inc. - Remote
***BI Developer**, July 2024-Present*

- Engineered and deployed HistGradientBoosting classifier for product profitability scoring using revenue, cost, pricing, and location data to achieve 91% precision and 89% recall.
- Designed, trained, and deployed production Propensity Lead Classifier; drove >$4M annual revenue with F1-score outperforming two third-party vendors.
- Mentored intern on NLP sentiment classifier for service tickets; automated analysis and reduced manual review effort by 50%+ across ~5,000 tickets/month.
- Led API integration pipeline for 22 MW contracted capacity tracking; enabled accurate forecasting and strategic planning.
- Co-authored Data Science Charter with SVP; automated deployment pipelines (Boomi) and streamlined Azure project tracking.

Nolan Transportation Group, Inc. (NTG) - Hybrid
***Data Science Analyst**, November 2022 - April 2024*

- Inherited and productionized outsourced classifier for lane reliability; optimized revenue on >60% of bids.
- Built ARIMA forecasting model for monthly linehaul rates; achieved 1.9% MAPE and informed future classifier logic.
- Developed calculated pricing field; A/B testing reduced MAPE by 1.6%.
- Created Django website + multi-source ETL pipeline feeding C-level metrics for $139M revenue portfolio.

***Pricing BI Analyst,** 2021 - November 2022*

- Automated JIRA task creation via REST API + Outlook parsing; saved ~$50K/year in labor.
- Built SSRS/Tableau reports and Excel macros for pricing automation and geographic lane analysis.

Projects

AWS DeepRacer PPO Agent - Georgia Institute of Technology, OMSA - DEC 2025

- Designed and trained a stabilized PPO policy (CNN + LIDAR encoders, 1.7M parameters) across Time-Trial, Object-Avoidance, and Head-to-Head scenarios on reInvent2019-wide, reInvent2019, and Vegas tracks; eliminated 6+ crawling/wall-hugging exploits via capped, non-multiplicative reward function.
- Achieved consistent lap completion (sub-16s potential) after 75k+ episodes; transferred base policy successfully to object-avoidance and head-to-head variants.
- Containerized entire training pipeline in Docker on Ubuntu, enabling reproducible rollouts and efficient cloud-scale experimentation.

Overcooked Environment with MARL QMIX Architecture - Georgia Institute of Technology, OMSA - DEC 2025

- Designed and trained two cooperative QMIX agents with per-agent GRUs and a monotonic mixing network to solve onion-soup delivery across three increasingly complex kitchen layouts (cramped room, coordination ring, counter-circuit).
- Engineered dense + event-based reward shaping (onion/pot/dish guidance, idle penalties, collision avoidance, exploit killers) that improved initial convergence by 2.3 soups; implemented episodic memory dictionary to prevent reward gaming and double-charging bugs.
- Conducted ablation studies and 150-episode evaluations; demonstrated emergent coordination (Agent 0 preemptively fetching dishes) despite asymmetric observations.

Education

Georgia Institute of Technology: Master of Science in Analytics — Expected May 2026
Kennesaw State University: BS in Systems and Industrial Engineering — May 2018